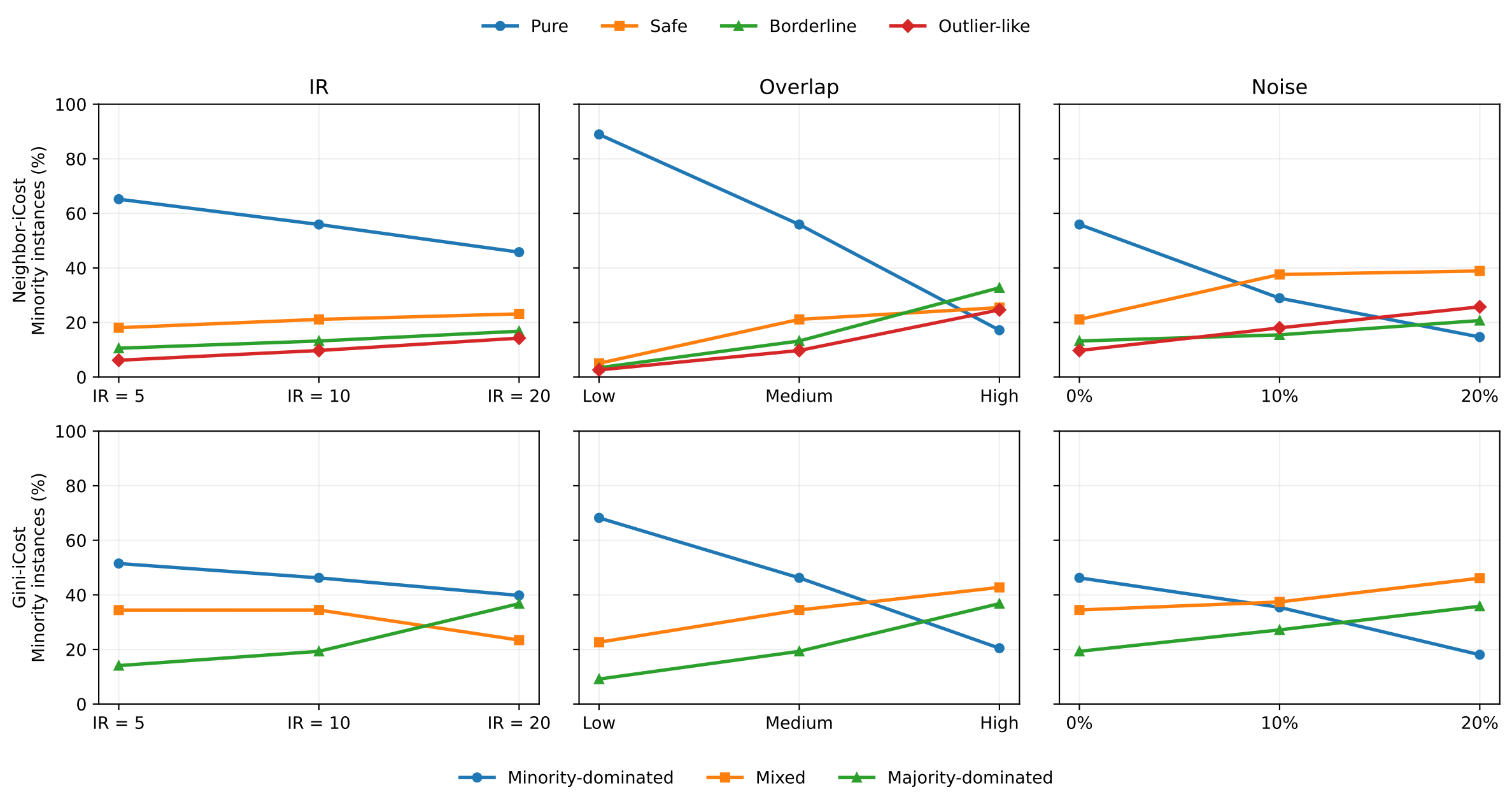
Response of Complexity Estimators to Controlled Data Difficulty

We examined whether the complexity estimates produced by the two iCost variants respond systematically to controlled changes in data difficulty. The five synthetic dataset families were used for this analysis. Starting from the reference condition (IR = 10, medium overlap, and no label noise) - IR, class overlap, and training-label noise were varied independently. The following figure shows the resulting distribution of minority instances across the complexity categories of Neighbor-iCost and the regional types of Gini-iCost.

For Neighbor-iCost, increasing each source of difficulty shifts minority observations away from the easier categories. As IR increases from 5 to 20, the proportion of pure minority instances decreases from approximately 65.2\% to 45.8\%, while the proportion of outlier-like instances increases from 6.2\% to 14.3\%. The effect is considerably stronger for class overlap. From low to high overlap, the pure proportion decreases from 88.9\% to 17.2\%, whereas borderline and outlier-like instances increase from 3.4\% to 32.7\% and from 2.6\% to 24.6\%, respectively. Training-label noise produces a similar redistribution: increasing noise from 0\% to 20\% reduces the pure proportion from 55.9\% to 14.7\% and increases the outlier-like proportion from 9.7\% to 25.7\%. These trends indicate that the neighborhood-based estimator responds directly to the loss of local minority support caused by increasing imbalance, overlap, and contamination.

Gini-iCost exhibits a similar regional response. With increasing overlap, noise, and IR, the proportion of minority observations located in minority-dominated regions decreases, while the proportions in mixed and majority-dominated regions generally increase. Overall, both estimators respond systematically to controlled changes in data difficulty, supporting their use for distinguishing heterogeneous minority-instance structures rather than assigning a uniform complexity level to the entire minority class.


Pure
Safe
Borderline
Outlier-like
IR
Overlap
Noise
Neighbor-iCost
Minority instances (%)
Gini-iCost
Minority instances (%)
100
80
60
40
20
0
IR = 5
IR = 10
IR = 20
Low
Medium
High
0%
10%
20%
Minority-dominated
Mixed
Majority-dominated